CSA-Net: An Adaptive Binary Neural Network and Application on Remote Sensing Image Classification

Weifeng Gao, Menghao Tan, Hong Li, Jin Xie, *Member, IEEE*, Xiaoli Gao, and Maoguo Gong, *Fellow, IEEE*

***Abstract*— When deep neural networks are used to process remote sensing data, complex network structures, and parameters limit their real-time processing deployment in satellites. Binary Neural Networks (BNNs) can overcome this obstacle by reducing network complexity. However, the binarization process naturally leads to information loss, reducing the performance of the original neural networks. In this article, three modules are introduced to improve the performance of BNNs. First, we propose an improved channel shuffle (CS) module to enhance communication between different channels in feature maps, in which the grouped pointwise convolution is used to reduce the original calculation scale. Second, the squeeze and excitation (SE) module called the attention mechanism is inserted before each layer of the binary convolution operation to recognize the importance of different feature channels. Third, the adaptive linear scaling factor (ALSF) module is proposed to make the weights of the convolution kernel more suitable for the structure of BNNs, which further improves the performance of our network. With these three modules, our CSA-Net reaches or even outperforms the other state-of-the-art (SOTA) methods with a lower computational cost. It can achieve 65.35% Top-1 accuracy on the ImageNet, 90.11% accuracy on the CIFAR10, 89.58% accuracy on the WHU-RS19, and 90.86% accuracy on the UC-Merced, respectively. Meanwhile, compared with other mainstream remote sensing classification models, the number of our parameters is reduced by two or even three orders of magnitude, and the results confirm the effectiveness of the proposed method.**

***Index Terms*— Adaptive linear scale factor, attention mechanism, binary neural network (BNN), channel shuffle (CS).**

I. Introduction

With the rapid development of neural networks, the deployment of neural networks on mobile devices and edge computing platforms has received significant attention from both industry and academia, such as image classification [1], object detection [2], autonomous driving [3], and so on. In recent years, many researchers have applied them to remote sensing data processing. However, the challenge of addressing the computation and network requirements associated with deep neural networks is difficult, particularly when individuals need to implement deep neural networks on low-power platforms [4].

Recently, many studies have been devoted to the design of high-precision networks for remote sensing image classification [5], [6], [7], [8], [9]. However, these methods do not strike the right balance between the accuracy and lightness of the model. It is necessary to create a lightweight enough classification network for remote sensing images that can be easily deployed on resource-limited platforms.

Binary neural networks (BNNs) [14] have become a promising option in lightweight network architectures. The reduction in computation and storage requirements improves the efficiency and speed of the neural network. BNNs are widely used in computer vision [15], emotion recognition [16], and other fields. Compared to traditional floating-point calculations, BNNs improve efficiency on the same hardware and achieve fast inference in resource-constrained devices [17]. In the development of BNNs, researchers have proposed various binary methods and algorithms, including Binary-Net [18], XNOR-Net [19], and DoReFa-Net [20]. These methods aim to address the unique challenges and requirements of binary networks.

However, one of the major challenges hindering the development of BNNs is the obvious performance gap of the relevant full-precision network. Therefore, exploring a high-precision and efficient BNN is an interesting and challenging task. The study has shown that MobileNet [21] is a compact and lightweight network, and the binarized MobileNet is a suitable candidate architecture. In particular, the ReActNet [22] has improved upon MobileNet to become the state-of-the-art (SOTA) model in BNNs. The advantage enables us to adopt the network structure of MobileNet as a baseline.

The two identical input blocks are connected to double the number of channels in the down-sampling layer of ReActNet. The problem of repeated calculation brings the redundancy of the same information, and binarization brings the serious loss of the information. Therefore, a more effective structure is proposed, which is inspired by the ShuffleNet [23] to realize the information exchange of the feature map

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Weifeng Gao, Menghao Tan, Hong Li, and Jin Xie are with the School of Mathematics and Statistics, Xidian University, Xian 710126, China (e-mail: gaoweifeng2004@126.com; tanmenghao@stu.xidian.edu.cn; lihong@mail.xidian.edu.cn; xj6417@126.com).

Xiaoli Gao is with Sichuan Jiuzhou Electronic Group Company Ltd., Mianyang 621000, China (e-mail: xiaoligao22@163.com).

Maoguo Gong is with the Key Laboratory of Intelligent Perception and Image Understanding, International Research Center for Intelligent Perception and Computation, Ministry of Education, Xidian University, Xian 710071, China (e-mail: gong@ieee.org).

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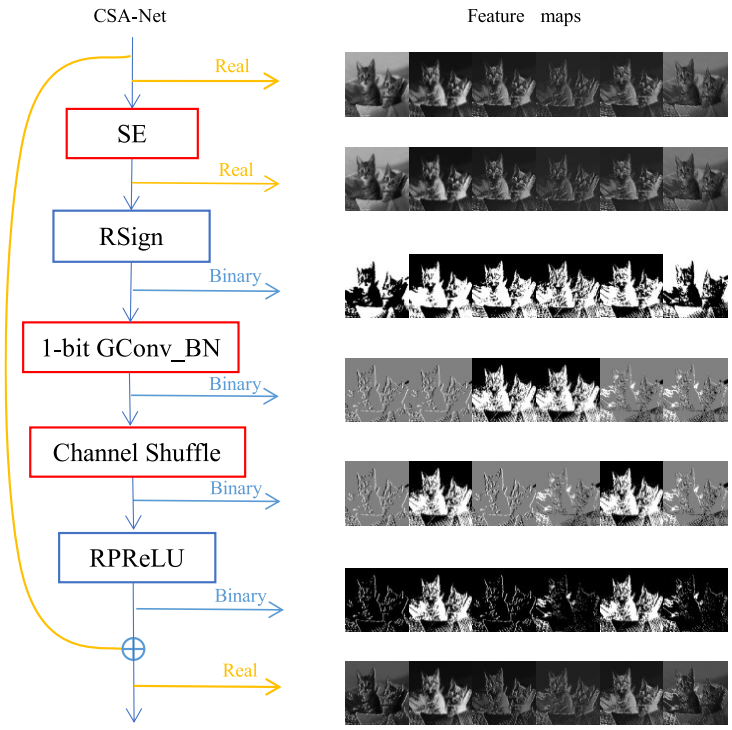


Fig. 1. CSA-Net with feature visualization after various modules. The blue line arrows represent binary activations propagated inside the path while the yellow line arrows represent real activations. The blue rectangles represent the original modules in the network structure while the red rectangles represent the modules that have been added and modified.

between different channels with low complexity. By using the channel shuffle (CS) module to improve the original structure, there is less redundant information and more valid information.

The attention mechanisms are being studied extensively, which usually are inserted in the first and penultimate layers. The activations after the binarized operations become $+1$ or -1 in the BNNs. The activations of the next layer input are the output of the valid information extracted by the previous layer, which means that it is important to extract more feature information. Deeper networks may result in fewer feature information being extracted after each layer due to binary operations (BOPs). Therefore, it is not useful to add the attention mechanism only in the first and penultimate layers. This article proposes to add the attention mechanism before the binary convolution of each layer to expect more feature information.

The XNOR-Net proposes a scaling factor of weight to fine-tune the convolution output. However, it is important to note that the fixed scaling factor is primarily aimed at minimizing errors of weight in both binary and full-precision networks, rather than exploring the optimal weight in binary convolution. This article proposes an adaptive linear scaling factor (ALSF) to calculate the optimal scaling factor of weight in the convolution kernel by back-propagation algorithm.

After the integration of CS, SE, and ALSF modules, the proposed network is called CSA-Net. The feature maps after various modules processing of the network are shown in Fig. 1, and the ALSF module exists in the binary convolution. It achieves a Top-1 accuracy of 65.35% on ImageNet, 90.11% on CIFAR-10, 89.58% on WHU-RS19, and 90.86% on UC-Merced, respectively. Moreover, the calculated scale of the CSA-Net is greatly reduced, surpassing other advanced BNNs. The experimental results validate the effectiveness of our proposed scheme, and our contributions are as follows:

1) We redesign the network architecture by adding the CS module with the grouped pointwise convolutions. Obtaining the rich feature map information in a more lightweight way improves the performance of the network.
2) The SE module is added before each layer of binary convolution. The channel attention mechanism enables binary convolution to focus on more effective information and improves network accuracy.
3) The ALSF module is proposed to replace the fixed scaling factor of weight, which can be updated by gradient descent rather than minimizing the quantification error between binary and full-precision convolution kernel.
4) We validate the effectiveness of the CSA-Net on CIFAR-10, ImageNet, WHU-RS19, and UC-Merced datasets. Extensive experimental results confirm the robustness and stability of the proposed method.

II. Related Work

As a pioneering work [18] of BNNs, the weights and activations are binarized by the Sign function, and the quantization errors are approximated by the straight-through estimator (STE). With further research, many improvements to BNNs have been developed to reduce the error. The

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development of BNNs can be described in the following three aspects [24], [25].

A. Optimization Method for the Binary Network

1) Minimize Quantization Error: The method is to reduce the quantization error of weights and activations in binary networks compared to full-precision networks. This involves ensuring that the quantization parameters closely match the full-precision parameters. Rastegari et al. [19] proposed XNOR-Net, an early study that addresses quantization error by binarizing both weights and activations. A unique aspect of this work is the introduction of a scaling factor to approximate floating-point weight. Similarly, XNOR-Net++ [26] combines weights and activations scaling factors into a single factor, enabling fine-tuning of the convolutional output.

2) Improve Network Loss Function: The method is to guide the learning of binarized parameters by designing specific network loss functions. Xue et al. [27] proposed to restrict the input information and recover feature information to enhance the capabilities of BNNs. Ye et al. [28] proposed using four relevant loss functions to train a lightweight and efficient BNN. There is also a knowledge transfer method, proposed by Leroux et al. [30], which uses a large-scale and full-precision teacher network to guide the training process of the binary student networks. Ding et al. [29] proposed using distribution loss to explicitly regularize the activations flow and devised a framework to methodically formulate the loss as follows:

$$L_{\text{total}} = L_{\text{CE}} + \lambda L_{\text{DL}} \tag{1}$$

where one of them is the cross-entropy loss for classification, and the other is the distribution loss for reducing the loss of binarization.

3) Reduce Gradient Error: The method is to design an approximate binarization function to help mitigate gradient mismatches in backpropagation. Liu et al. [31] proposed a custom ApproxSign function to replace the Sign function in the gradient meter used in Bi-Real Net. By designing an approximation function, they aim to improve the compatibility of the Sign function in the back-propagation process. Gong et al. [32] proposed a method called differentiable soft quantization (DSQ) with a soft quantization function

$$\phi(x) = s \tanh(k(x - m)) \tag{2}$$

where k, s, and m are the hyperparameters of the asymptotic function and determine the shape of the function. Qin et al. [33] used an adaptive error decay estimator (EDE) to gradually reduce the gradient error in training, and the EDE balances the ability to update weight and the goal of reducing gradient error. Then, the Distribution-sensitive Two-stage Estimator (DTE) [34] is proposed to retain the information of gradients by distribution-sensitive soft approximation.

B. Efficient Model Design

1) Binarize Manually Designed Networks: VGG [35] and ResNet [36] are widely recognized as benchmark models in the field of binarization due to their strong generalization performance. On the one hand, VGG [35] is known for its deep network structure comprising 16 or 19 layers, consisting of simple convolutional and pooling operations. Therefore, it is a common and simple method to use the VGG network as a binary benchmark network. On the other hand, ResNet [36] introduces the concept of residual learning, which effectively addresses the challenge of gradient vanishing and exploding in deep neural networks through cross-layer connections. With a deeper network structure compared to VGG [35], ResNet [36] has achieved remarkable performance in various image tasks.

These benchmark networks are often directly binarized in the expectation of obtaining binary networks with high performance. MobileNet [21] is a compact convolutional neural network known for its efficiency and accuracy, which makes it particularly suitable for deployments on mobile devices. By employing depthwise separable convolution, MobileNet maintains high accuracy while reducing model parameters and computational complexity. The study has shown that the network structure of MobileNet is more compact and better suited for training binary networks, and we benefit from using the network structure of MobileNet.

2) Neural Architecture Search: There is a lot of attention on neural architecture search (NAS) [37] because it can improve network performance and generalization by automatically searching for the optimal network architecture. The three main categories of NAS methods are as follows: evolution-based methods, reinforcement-based learning methods, and gradient-based methods [10]. Specifically, evolution-based methods [11], [12], [13] typically require a lot of computing resources and expertise to find high-performance encoding of network structures. Reinforcement-based learning methods typically require more computing resources, and even using thousands of graphics processing cards (GPUs) on medium-sized datasets can take days. Gradient-based methods are more efficient than evolution-based methods and reinforcement-based learning methods. However, due to the improper relationship of adaptation to gradient optimization, researchers often find pathological architectures.

Unfortunately, this relationship has not been mathematically proven. In addition, gradient-based methods require the construction of a super-net in advance, which also requires a high degree of expertise. Therefore, combining BNNs and NAS is a promising approach, but it will be more difficult.

C. Binarization Operations

The BOP of BNNs is achieved through the Sign function. The Sign function can binarize the weights and activations of CNN. ReActNet [22] shifts and even reshapes the distribution of activations, in which the RSign/RPReLU functions are used to replace the Sign/PReLU functions, and the functions are as follows:

$$\begin{aligned}
x_b &= \text{Sign}(x) = \begin{cases} +1, & \text{if } x \geq 0 \\ -1, & \text{if } x < 0 \end{cases} \\
x_b &= \text{RSign}(x) = \begin{cases} +1, & \text{if } x \geq \alpha \\ -1, & \text{if } x < \alpha \end{cases} \\
x_b &= \text{RPReLU}(x) = \begin{cases} x - \alpha + \gamma, & \text{if } x \geq \alpha \\ \mu(x - \alpha) + \gamma, & \text{if } x < \alpha \end{cases}
\end{aligned} \tag{3}$$

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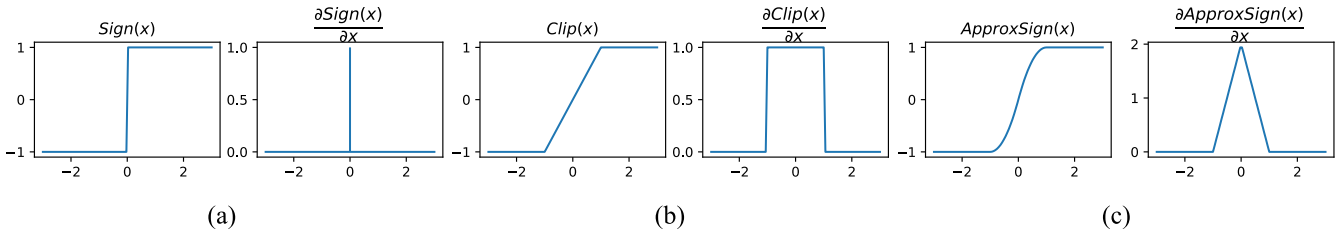


Fig. 2. (a) Sign function and its derivative. (b) STE and its derivative for replacing the derivative of the Sign function in gradient computation, proposed by Courbariaux et al. [18]. (c) Quadratic piecewise polynomial function for replacing the derivative of the Sign function in gradient computation, proposed by Liu et al. [31].

where x denotes the real-valued activations, x_b denotes its binarized activations. μ represents a coefficient that can be trained to influence the slope of the negative semiaxis. α and γ are adjustable shifts used to reshape the distribution along the horizontal and vertical axes. The RSign and RPReLU functions are used in our CSA-Net.

It is impossible to update the parameters by directly applying the gradient descent algorithm in the training process due to the nondifferentiable nature of binary functions, such as the Sign and RSign functions. The derivative of the Sign function is 0 almost everywhere, which poses a challenge to update binary weight in the back-propagation process. The STE is used in the gradient descent algorithm to resolve the challenge. By applying STE, the BNNs can be trained using the same gradient descent algorithm as full-precision neural networks. At present, the mainstream method still uses the STE to train binary networks. In practice, some derivable functions are also used to replace the Sign function, and the quadratic piecewise polynomial function is used to approximate the Sign function in Bi-Real Net [31]. The functions are shown in Fig. 2. At the same time, this is also a simpler and more universal method in binary networks, so we also use this method to approximate the Sign function in our CSA-Net. The ApproxSign function $F(x)$ is proposed as follows:

$$F(x) = \begin{cases} -1, & \text{if } x < -1 \\ 2x + x^2, & \text{if } -1 \le x < 0 \\ 2x - x^2, & \text{if } 0 \le x < 1 \\ 1, & \text{otherwise} \end{cases}$$

$$\frac{\partial F(x)}{\partial x} = \begin{cases} 2 + 2x, & \text{if } -1 \le x < 0 \\ x - 2x, & \text{if } 0 \le x < 1 \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

III. Proposed Algorithm

This article aims to design a high-precision and efficient BNN called CAS-Net. Specifically, we introduce the CS module of grouped pointwise convolution in Section III-A, which is to get richer feature information while reducing the computational cost. Furthermore, we explore how to use the plug-and-play Squeeze and Excitation (SE) module called the attention mechanism to improve performance in Section III-B. Finally, the mathematical process and role of the ALSF module are discussed in Section III-C.

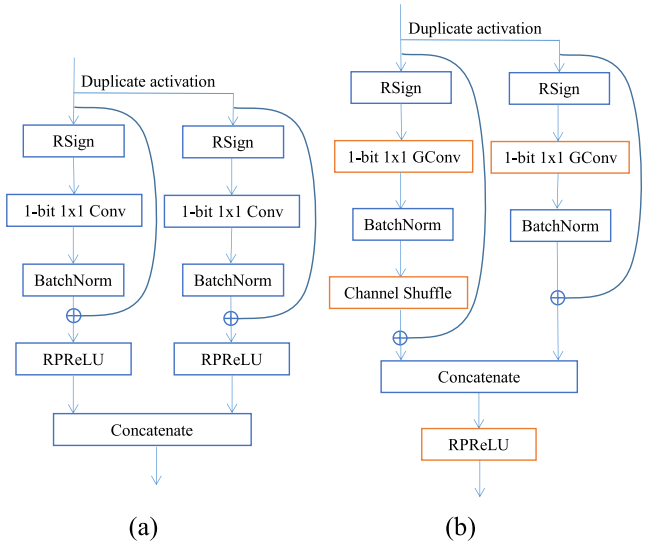


Fig. 3. Design of the down-sampling layer structure. (a) Original binary convolutional block in ReActNet. (b) Improved binary convolutional block in CAS-Net.

A. CS Module

In the down-sampling layer structure, there are repeated operations with a large number of pointwise convolution parameters in ReActNet, as shown in Fig. 3(a). The structure inevitably increases computation redundancy but does not increase the amount of feature information. The primary objective of building high-performance binary networks is to reduce the number of model parameters and computation redundancy without compromising accuracy.

Inspired by ShuffleNet [23], we propose an improved down-sampling layer structure that reduces computation and increases the exchange of information between different channels, as shown in Fig. 3(b). First, the two pointwise convolutions are replaced by the grouped pointwise convolution, and the number of parameters and computations is reduced to the grouped times of the original convolutions. Then, one of the grouped pointwise convolution operations is followed by the CS module for effective fusion and communication between channels by shuffling their order. The other grouped pointwise convolution operation is left without additional processing to retain more information from the original channel. Remote sensing images are more difficult to recognize and generally contain more complex backgrounds and rich textures. Our improved structure provides richer feature information and lowers computational cost compared to

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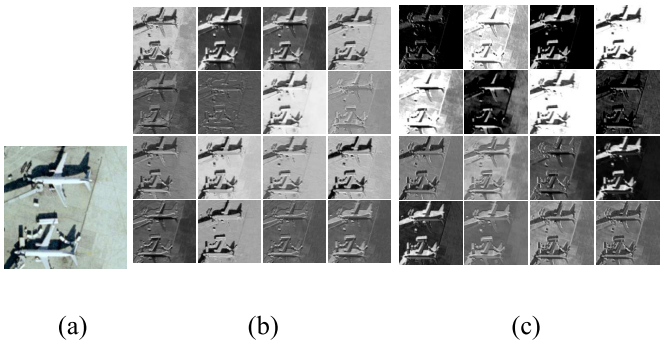


Fig. 4. (a) Remote sensing image of the aircraft. (b) Feature map after the original pointwise convolution. (c) Feature map after the grouped pointwise convolution and CS module.

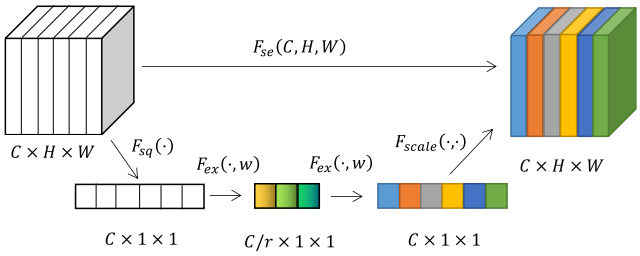


Fig. 5. Specific process of the operation of the channel attention mechanism.

Fig. 3(a). This structure is more effective for feature extraction of remote sensing images.

To better explore the role of the CS module on remote sensing images, some feature maps after the convolution are visualized, as shown in Fig. 4. The original structure can result in the feature mapping of multiple pairs of duplicates, as shown in Fig. 4(b). Although rich feature maps may be an essential feature of a better-performing network, the repeated operation of extracting features makes the binarized image information more ambiguous. After deploying the CS module, the information on the feature maps is more diverse, as shown in Fig. 4(c), which proves that the CS module works well.

B. SE Module

The SE module is designed to capture the interdependence of feature channels. It is achieved by using the fully connected (FC) neural networks to automatically learn the importance of each feature channel. Then, each feature channel is assigned a corresponding weight based on its importance, which allows the model to focus on the most important channels for the current task. Therefore, The SE module is called the channel attention mechanism, whose specific details are shown in Fig. 5. The SE module enhances the useful feature channels and suppresses unimportant feature channels.

The "squeeze" part of the SE module aims to compress the original feature map dimensions from number of channels, height, and width (CHW) to C × 1 × 1. This compression is achieved by applying a global average pooling operation, which effectively captures the global visual field of the feature map and provides a broader perspective.

The "excitation" part of the SE module sends the C × 1 × 1 representation obtained to the FC layer. The FC layer predicts

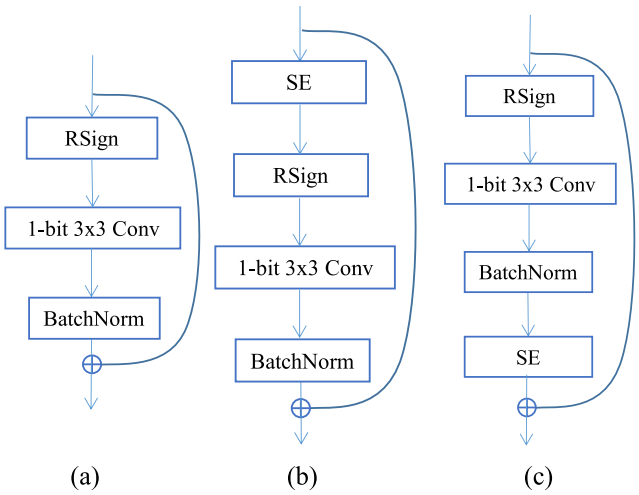


Fig. 6. Structural design with different locations of the SE module. (a) Normal structure. (b) Structure with the Previous-SE module. (c) Structure with the Standard-SE module.

the importance of each feature channel and rewards the corresponding original feature channels by linear transformation. This redistribution of feature channels based on their importance enhances the sensitivity of the neural network to relevant channel features.

The target objects in remote sensing images are often smaller and need to be paid more attention. At the same time, the SE module can help focus on needed areas of image and the module can be easily integrated into existing network structures. Since binarization can result in significant information loss, it is more important to focus on the useful feature information of different channels in each layer in remote sensing images. The SE module has different effects on extracting features at different locations of the network structure. There are two classic methods to insert an SE module in a network structure. One method is to insert in front of the convolution called the Previous-SE module, and the other is to insert in the back of the convolution called the Standard-SE module, as shown in Fig. 6. The impact of two SE module structures on our CSA-Net is explored in the ablation study.

C. ALSF Module

XNOR-Net first uses the fixed scaling factor to minimize the error of weight between binary and full-precision networks as follows:

$$\min_{\alpha,b_w} \|w - \alpha b_w\|^2 \tag{5}$$

where w is the weight of the full-precision convolution kernel, b_w is the binarized weight, and α is the scaling factor. The use of the scaling factor can accelerate training and improve accuracy to some extent and scholars have taken the operation as a standard method of binary network optimization. However, it is not confirmed that the use of the fixed scaling factor is the most suitable operation for BNNs. Among them, Bai et al. [53] pointed out that the fixed scaling factor does not significantly enhance the performance of BNNs and can be

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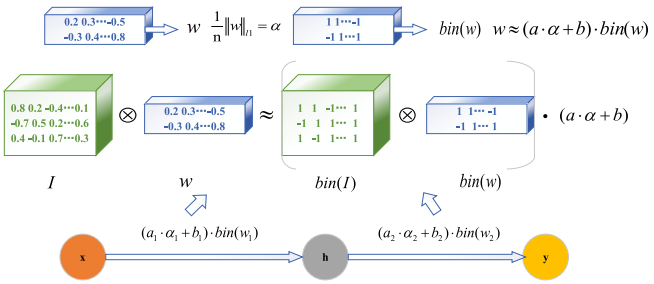


Fig. 7. Specific process of binarization weights and activations in the forward propagation process, and the scale factor α is replaced by the adaptive linear scale factor $a \cdot \alpha + b$.

seen as redundant work, as there is no substantial contribution to the model. In this article, the ALSF is used to automatically adjust through the back-propagation algorithm instead of the original fixed scaling factor. The ALSF is given as follows:

$$w \sim \alpha \cdot \text{bin}(w) \sim (a \cdot \alpha + b) \cdot \text{bin}(w). \tag{6}$$

When a and b are adjustable parameters, a initializes to close to 1 and b initializes to close to 0 for ensuring the stability of the initial training.

The origin weight $\alpha \cdot \text{bin}(w)$ is replaced by the improved weight $(a \cdot \alpha + b) \cdot \text{bin}(w)$, and the calculation of binarization weights and activations in the forward propagation process, as shown in Fig. 7. There is no doubt that the linear adaptive scale factor $a \cdot \alpha + b$ can be updated by gradient descent. Here is a three-layer example in the network to explain how to update parameters by the back-propagation algorithm. As shown in Fig. 7, x is the input node, h is the intermediate node and y is the output node. The learning goal of the CSA-Net is to find w, a, and b. We assume E is the loss of the output node, and the back-propagation process is as follows:

$$w_1^* = w_1 - \eta \cdot \frac{\partial E}{\partial w_1}, \quad a_1^* = a_1 - \eta \cdot \frac{\partial E}{\partial a_1}, \quad b_1^* = b_1 - \eta \cdot \frac{\partial E}{\partial b_1}. \tag{7}$$

The partial derivative of the loss E to variables can be developed by the chain rule, as

$$\begin{aligned} \frac{\partial E}{\partial w_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial w_1} \\ \frac{\partial E}{\partial a_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial a_1} \\ \frac{\partial E}{\partial b_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial b_1} \end{aligned} \tag{8}$$

while the partial derivative of the output y to variables can also be developed by the chain rule, as

$$\begin{aligned} \frac{\partial y}{\partial h} &= (a_2 \cdot \alpha_2 + b_2) \cdot \text{bin}(w_2) \\ \frac{\partial h}{\partial w_1} &= x \cdot (a_1 \cdot \alpha_1 + b_1) \cdot \frac{\partial \text{bin}(w_1)}{w_1} \\ \frac{\partial h}{\partial a_1} &= x \cdot \alpha_1 \cdot \text{bin}(w_1), \quad \frac{\partial h}{\partial b_1} = x \cdot \text{bin}(w_1). \end{aligned} \tag{9}$$

It is important to note that $\text{bin}(w_1)$ is not differentiable, so the ApproxSign function $F(w_1)$ is used to update the gradient. Substituting (8) and (9) into (7) we have

$$\begin{aligned} \frac{\partial E}{\partial w_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial w_1} \\ &= \frac{\partial E}{\partial y} \cdot (a_2 \cdot \alpha_2 + b_2) \cdot \text{bin}(w_2) \\ &\quad \cdot x \cdot (a_1 \cdot \alpha_1 + b_1) \cdot \frac{\partial F(w_1)}{w_1} \\ \frac{\partial E}{\partial a_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial a_1} \\ &= \frac{\partial E}{\partial y} \cdot (a_2 \cdot \alpha_2 + b_2) \cdot \text{bin}(w_2) \cdot x \cdot \alpha_1 \cdot \text{bin}(w_1) \\ \frac{\partial E}{\partial b_1} &= \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial h} \cdot \frac{\partial h}{\partial b_1} \\ &= \frac{\partial E}{\partial y} \cdot (a_2 \cdot \alpha_2 + b_2) \cdot \text{bin}(w_2) \cdot x \cdot \text{bin}(w_1). \end{aligned} \tag{10}$$

The ApproxSign function is used to approximate the Sign function in the back-propagation process, and we keep the real numerical weights for gradient updates according to (10) during the training process. The method of ALSF is not only simple to guide but also improves the accuracy without increasing quite the calculation amount.

IV. Experiments

We perform image classification experiments on the CIFAR10 [38], ImageNet [39], WHU-RS19 [50], and UC-Merced [51] datasets to search for the performance of CSA-Net. The datasets and implementation details are introduced in Section IV-A. Then we describe the ablation study in Section IV-B, which explores the effects of the CS module, the SE module, and the ALSF module on network performance. Finally, in Section IV-C we compare the CSA-Net with other networks on different datasets in terms of accuracy and computational cost.

A. Datasets and Implementation Details

The CIFAR10 is a small dataset for identifying universal objects, which contains 60K images with a spatial resolution of 32 × 32. The training set and validating set are comprised of 50K images and 10K images, respectively. The images are all evenly distributed in ten categories common in life.

The ImageNet is extensive with more diversity, which consists of 1.3 million training images and 50K test images. The images are the high resolution of different sizes and roughly evenly distributed across 1K categories.

To ensure experiment comparability, we apply data augmentation methods as in prior binary network research [19]. For the CIFAR10 dataset, random cropping, image lighting, and random horizontal flipping are adopted. For the ImageNet dataset, data augmentation and label smoothing [31] are added.

The WHU-RS19 dataset contains about 1K images with a size of 600 × 600 pixels, which is composed of 19 classes of different scenes. The images used in this study are sourced from Google Earth imagery. As per the methodology outlined in reference [50], 50% of the images within each category are chosen at random to comprise the training set.

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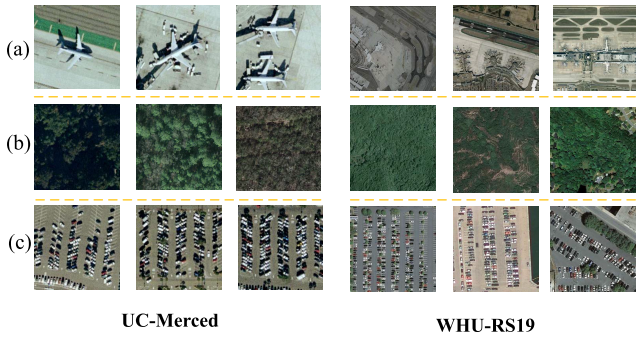


Fig. 8. Samples of remote sensing scenes on the different datasets. (a) Airport. (b) Forest. (c) Parking lot.

The UC-Merced dataset contains 2.1K images with a size of 256 × 256 pixels, which is composed of 21 classes. The images used in this study are sourced from the USGS National Map. As per the methodology outlined in reference [51], 50% of the images within each category are also chosen at random to comprise the training set.

The remote sensing images depict various land-cover units of varying sizes, as shown in Fig. 8. In remote sensing classification, the overall accuracy and the average accuracy are widely evaluation measures. The overall accuracy represents the proportion of correctly classified images to the total number of validating images. To be consistent with the evaluation criteria of the universal dataset, we uniformly use the overall accuracy as the evaluation measure.

The networks are optimized by using Adam. During the training on the CIFAR10 and remote sensing datasets, the initial learning rate is set to 1e-3, weight decay to 1e-4, and momentum to $\beta = (0.9, 0.999)$. During the training on the ImageNet dataset, the initial learning rate is set to 5e-4, weight decay to 3e-5, and momentum remains the default. To further speed up convergence, the learning rate is reduced according to the 0.1 step every 50 epochs in the ablation study. All experiments are implemented in PyTorch of Windows 10, Intel Core i7-11700, NVIDIA RTX 3090 GPU, and 64 GB RAM.

B. Ablation Study

In this section, we will provide further evidence of the effectiveness of our proposed modules. The original ReActNet is only proposed for the ImageNet dataset design. For images with small resolution in the CIFAR10 dataset, excessive down-sampling will lead to information loss too much. Therefore, the stride of the first real-value down-sampling convolution is set to 1 in the original network, and the new baseline is called "Vanilla." The Vanilla network is the baseline network based on improvements of ReActNet. Then, our ablation studies are conducted on the CIFAR10 dataset with 300 epochs. The validating results of the Vanilla network of the different combinations with the CS module, SE module, and ALSF module, as shown in Table I. The results show that our proposed methods can achieve higher accuracy, as shown in Fig. 9.

1) CS Module: The CS module can further reduce the computational complexity of the BNNs. The ReActNet reproduces the down-sampling architecture and connects outputs, as shown in Fig. 3(a). However, repeated operations make no sense in the growth of feature information for neural networks. The Vanilla network is unstable compared to the original ReActNet network in training, as shown in Fig. 10. However, in the end, it gets better performance which also indicates that our modification of the down-sampling operation is effective on the CIFAR10 dataset. For the training of unstable and operational redundancy, we propose the CS module to solve this problem, in Fig. 3(b). Only one of the grouped pointwise convolution operations is followed by the CS module to avoid repeated information redundancy and implement the different channel information fusion. The part will be connected with the other grouped pointwise convolution operation to keep the channel aligned. This operation also retains more of the feature information of the previous channels.

TABLE I
ABLATION STUDIES OF THE CS MODULE, SE MODULE, AND ALSF MODULE WITH THE VANILLA NETWORK COMBINATION, RESPECTIVELY

Settings	Top-1 accuracy(%)
Vanilla	88.46
Vanilla+CS	89.48
Vanilla+SE	89.65
Vanilla+ALSF	89.64
Vanilla+CS+SE	90.07
Vanilla+CS+ALSF	90.11
Vanilla+SE+ALSF	88.92
Vanilla+CS+SE+ALSF	90.17

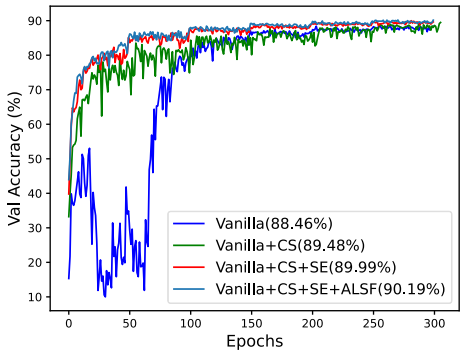


Fig. 9. Validating results of the CS module, SE module, and ALSF module combined with the Vanilla network.

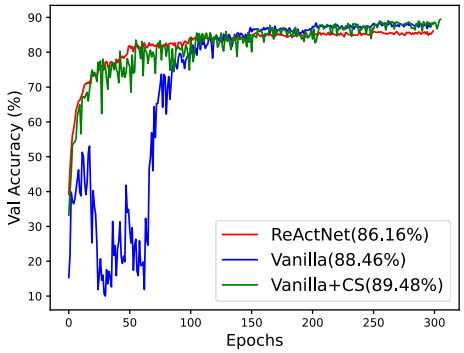
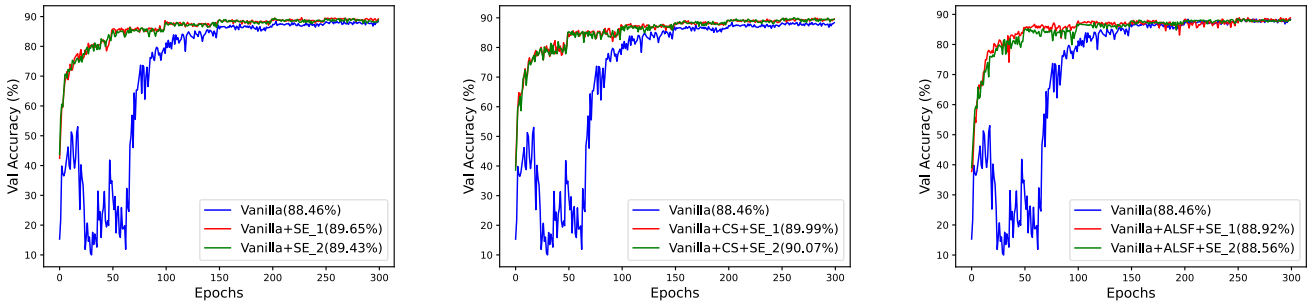


Fig. 10. Ablation Studies of the CS module. The validating results of the ReActNet, the Vanilla network, and "Vanilla + CS."

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(a) (b) (c)

Fig. 11. Ablation Studies of the SE module. (a) Validating results of the Vanilla network and "Vanilla + SE." (b) Validating results of the Vanilla network and "Vanilla + CS + SE." (c) Validating results of the Vanilla network and "Vanilla + ALSF + SE."

TABLE II

COMPARISON OF ERROR AND PARAMETERS WITH SOTA NETWORKS ON CIFAR10. THE FOLLOWING COMPARISONS ARE FOR NETWORKS WITH BOTH ACTIVATIONS AND WEIGHTS BINARIZED

Binary Networks	Test Error (%)	Parameters (M)
ResNet18[41]	14.5	0.59
WRN-22[42]	11.48	4.33
PC-DARTS[43]	8.94	4.2
XNOR-Net[19]	10.2	3.4
DoReFa-Net[20]	10.0	3.7
IR^2-Net[27]	13.7	-
DSQ[33]	8.3	-
DIR-Net[34]	11.0	-
ReActNet[22]	13.84	0.29
Vanilla	**11.54**	**0.29**
Ours	**9.89**	**0.33**

TABLE III

COMPARISON OF PARAMETERS, COMPUTATIONAL COST, AND MEMORY USAGE FOR THE VANILLA NETWORK WITH DIFFERENT MODULES

Binary Networks	Parameters (M)	OPs $\times 10^6$	FLOPs $\times 10^6$	BOPs $\times 10^7$	Memory Usage(M)
Vanilla	0.29	6.25	0.11	39.32	23.65
Vanilla+CS	0.29	5.58	0.11	34.94	29.65
Vanilla+CS +SE	0.33	5.58	0.12	34.94	31.71
Vanilla+CS +SE+ALSF	0.33	5.58	0.12	34.94	31.71

2) SE Module: We further evaluate the impact of the attention mechanism on binary structure. As mentioned in Section III, we provide an ablation study of the two different insertion structures with the Standard-SE module and the Previous-SE module. The results of combining these two SE modules with other modules are shown in Fig. 11. The attention mechanism helps to alleviate the problem of information loss and further enhances the role of the CS module in BNNs.

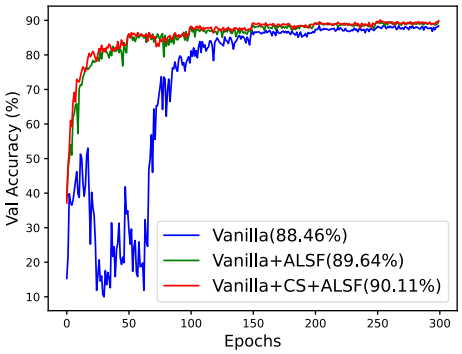


Fig. 12. Ablation Studies of the ALSF module. The validating results of the Vanilla network, "Vanilla + ALSF," and "Vanilla + CS + ALSF."

The results show that the Previous-SE module, which adds the attention mechanism before the convolution, improves the Vanilla network better because the binarization of each layer will lead to a loss of information in the feature graph. The attention mechanism added before convolution can help the network to focus on some feature channels, and the results meet our predictions.

However, for the CS module, the effect of the attention mechanism added after convolution is slightly better than that before convolution. We think that due to the CS module, the feature information of the channels has become more sufficient. The subsequent attention mechanism further plays the role of the CS module.

3) ALSF Module: In previous research on binary networks, the scaling factor minimizes the error of weight between the binary and the full-precision convolution kernel. However, this article believes that the different scaling factors may achieve better results. Since the original scaling factors can ensure the normal operation of binary training, we use the linear scaling factor to fine-tune the scaling factor. The ALSF of design not only ensures the continuity of training but also conforms to the automatic back-propagation algorithm. We have achieved good results without increasing the amount of computation. The results of the combination of ALSF and other modules are shown in Fig. 12.

C. Results

We evaluate the experimental results of the proposed method and compare the results with the SOTA BNNs in this

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TABLE IV
COMPARISON OF ACCURACY, COMPUTATIONAL COST, AND PARAMETERS WITH SOTA BINARY NETWORKS ON IMAGENET. THE FOLLOWING COMPARISONS ARE FOR NETWORKS WITH BOTH ACTIVATIONS AND WEIGHTS BINARIZED. * REPRESENTS THE MORE COMPLEX NETWORKS FROM THE ORIGINAL NETWORKS

Binary Networks	Accuracy (%)		Parameters	OPs	FLOPs	BOPs	Method
	Top1	Top5	(M)	$\times 10^8$	$\times 10^8$	$\times 10^9$	
Binary-Net[18]	42.2	-	-	1.47	1.20	1.70	Manual
XNOR-Net[19]	51.2	73.2	3.9	1.67	1.41	1.70	Manual
Bi-Real-Net-18[31]	56.4	79.5	11.8	1.63	1.39	1.68	Manual
Bi-Real-Net-34* [31]	62.2	83.9	22.9	1.93	1.39	3.53	Manual
IR^2-Net[27]	63.8	85.5	-	1.74	1.48	1.68	Manual
PCNN[48]	57.3	80.0	-	-	1.67	-	Manual
MeliusNet29[40]	65.8	86.2	1.9	2.14	1.29	5.47	Manual
MeliusNet42*[40]	69.2	88.3	3.8	3.25	1.74	9.69	Manual
RBCN[49]	59.5	81.6	3.9	-	1.67	-	Manual
ReActNet-A[22]	69.4	89.5	4.4	0.87	0.12	4.82	Manual
ReActNet-B*[22]	70.1	-	8.1	1.63	0.44	4.69	Manual
DIR-Net[34]	63.3	85.1	-	-	-	-	Searched
BNAS-H[44]	63.51	83.91	-	4.71	3.73	6.28	Searched
BATS[45]	60.4	83.0	-	0.99	0.81	1.15	Searched
BNAS[46]	64.3	86.1	6.4	-	-	-	Searched
EBNAS[47]	62.6	84.2	-	1.02	0.83	1.23	Searched
Ours	**65.35**	**85.74**	**4.2**	**0.79**	**0.12**	**4.28**	**Manual**
Ours*	**66.28**	**86.46**	**6.86**	**2.01**	**1.34**	**4.28**	**Manual**

section. To quantify the computational load, the BOPs and floating-point operations (FLOPs) are separately counted, and we can obtain the total operations (OPs) by using the formula OPs = BOPs + FLOPs [22].

The evaluation results of the CIFAR10 dataset are shown in Table II. The CSA-Net achieves better accuracy on CIFAR10 compared to manually designed SOTA BNNs. Moreover, compared with ReActNet, our network improves the model accuracy by 4.58% with a similar number of parameters. Although the accuracy of CSA-Net is 0.95% less than PC-DARTS of NAS, our number of parameters is one-tenth of PC-DARTS. As a result, our CSA-Net has demonstrated strong performance on the small dataset CIFAR10. We further compare the impact of different modules on the network, as shown in Table III. The results also validate the characteristics of the proposed modules. For example, the CS module reduces the computational cost of BOPs, and the SE and ALSF modules improve accuracy without adding much computation, even though the SE module increases a small number of parameters.

While we also evaluate the CSA-Net on the ImageNet dataset, the comprehensive results are presented in Table IV. We observe the CSA-Net has fewer OPs, FLOPs, and BOPs than almost all BNNs, and the accuracy exceeds most of the searched SOTA BNNs. Furthermore, the accuracy of CSA-Net improves the manually designed SOTA Bi-Real-Net-34(62.2%) by 3.15% with only one-fifth of the original number of parameters. Compared with ReActNet, there is still a gap in the performance of our model, but the amount of computation in our model is greatly reduced.

Finally, we evaluate the CSA-Net on the remote sensing datasets. The classification results of the WHU-RS19 and UC-Merced datasets are shown in Table V. Our CSA-Net reduces the number of parameters while maintaining accuracy.

TABLE V
COMPARISON OF PARAMETERS AND OVERALL ACCURACY FOR DIFFERENT NETWORKS ON REMOTE SENSING DATASETS

	WHU-RS19(%)	UC-Merced(%)	Parameters (M)
VGG16[35]	89.74	93.56	138
VGG-VD-16[52]	96.05	94.14	138.36
MF^{2}CNet[53]	-	98.76	30.12
RSFJR[54]	97.48	97.21	550.98
Wavelet CNN[55]	86.2	92.19	10.24
LCNN-BFF[56]	-	94.64	6
ResNet18[57]	-	97.43	11.19
DCA-Fusion[5]	93.56	93.84	-
GLM16[6]	92.99	92.60	-
ViT[7]	-	98.48	49
EMTCAL[7]	97.6	98.67	27.0
SF-MSFormer[8]	97.8	99.35	26.12
LGLFformer[9]	99.2	99.48	22.96
Ours	**89.58**	**90.86**	**0.33**

Although there is an accuracy gap between our CSA-Net and some of the SOTA models, there is a hundredfold or even thousandfold difference in the number of parameters between them. It is helpful for the timely processing of remote sensing images and the deployment of deep neural networks. On our devices, the testing time for both the validation set and the per image is shown in Table VI. Due to the parallel operations of binarization and grouped convolutions, the inference time may be longer. The binary networks are only deployed on hardware, and the BOPs are converted to XNOR operations to improve the inference speed greatly. It should be noted that the CSA-Net is a binary network, and the network will bring great advantages in terms of storage and inference. At the

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TABLE VI
TESTING TIME PER IMAGE AND VALIDATION SETS ON DIFFERENT DATASETS BY DIFFERENT NETWORKS

	VGG		Vanilla		CSA-Net	
	Val. (s)	Per. (ms)	Val. (s)	Per. (ms)	Val. (s)	Per. (ms)
CIFAR-10	16.75	1.68	17.66	1.77	18.43	1.84
WHU-RS19	17.02	33.57	20.78	43.75	22.33	47.01
UC-Merced	18.48	17.60	25.71	24.49	27.47	26.16

same time, it also proves that the application of the binary network in remote sensing image classification is a promising practice.

Therefore, the improved method proposed in this article can further reduce the computation cost of binary networks and stably improve the performance of the model on small datasets. Our CSA-Net has greater computational efficiency when considering an equivalent parameter count in small datasets. At the same time, less computation will lead to the degradation of model performance on large datasets. In conclusion, our method results in a lighter and higher accuracy network structure, which enables a trade-off between model complexity and performance when faced with specific problems.

V. CONCLUSION

In this article, we present a novel BNN called CSA-Net for universal and remote sensing image classification. We first introduce the combination of the CS module and the grouped pointwise convolution, which enables effective information exchange between different channels. The accuracy of the Vanilla + CS on the CIFAR10 is improved by 1.02%, using only 88.86% of the original computational BOPs. Then, we integrate the SE module into the feature extraction layer structure to emphasize important features and suppress irrelevant ones. Lastly, we propose the ALSF module to fine-tune the weights of the binary convolution, which further improves the performance of the network without increasing the amount of computation. Based on those three methods, 65.35% Top-1 accuracy on the ImageNet, 90.11% accuracy on the CIFAR10, 89.58% accuracy on the WHU-RS19, and 90.86% accuracy on the UC-Merced are achieved, respectively, with a lower computational cost. Our next research direction is to balance the relationship between large remote sensing datasets and BNNs. We believe that it is greatly promising for BNNs on low-power platforms to process remote sensing data in the near future.

REFERENCES

[1] W. Rawat and Z. Wang, "Deep convolutional neural networks for image classification: A comprehensive review," *Neural Comput.*, vol. 29, no. 9, pp. 2352–2449, Sep. 2017.

[2] C. Papageorgiou and T. Poggio, "A trainable system for object detection," *Int. J. Comput. Vis.*, vol. 38, no. 1, pp. 15–33, 2000.

[3] L. Claussmann, M. Revilloud, D. Gruyer, and S. Glaser, "A review of motion planning for highway autonomous driving," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 5, pp. 1826–1848, May 2020, doi: 10.1109/TITS.2019.2913998.

[4] K. Yan, S. Mei, M. Ma, and F. Yan, "Fusing deep local and global features for remote sensing image scene classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Yokohama, Japan, Jul. 2019, pp. 3029–3032, doi: 10.1109/IGARSS.2019.8898963.

[5] S. Chaib, H. Liu, Y. Gu, and H. Yao, "Deep feature fusion for VHR remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4775–4784, Aug. 2017.

[6] Y. Yuan, J. Fang, X. Lu, and Y. Feng, "Remote sensing image scene classification using rearranged local features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1779–1792, Mar. 2019.

[7] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[8] Y. Yang et al., "An explainable spatial–frequency multiscale transformer for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5907515, doi: 10.1109/TGRS.2023.3265361.

[9] Y. Yang et al., "LGLFormer: Local-global lifting transformer for remote sensing scene parsing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602513, doi: 10.1109/TGRS.2023.3344116.

[10] M. Tan et al., "Universal binary neural networks design by improved differentiable neural architecture search," *IEEE Trans. Circuits Syst. Video Technol.*, doi: 10.1109/TCSVT.2024.3398691.

[11] Z. Wei, W. Gao, M. Gong, and G. G. Yen, "A bi-objective evolutionary algorithm for multimodal multi-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 28, no. 1, pp. 168–177, Feb. 2024, doi: 10.1109/TEVC.2022.3217258.

[12] W. Gao, L. Huang, S. Liu, F. T. S. Chan, C. Dai, and X. Shan, "Artificial bee colony algorithm with multiple search strategies," *Appl. Math. Comput.*, vol. 271, pp. 269–287, Nov. 2015.

[13] L. Zhang, W. Gao, J. Xie, and H. Li, "An ordinary differential equation for modeling halpern fixed-point algorithm," *Appl. Math. Lett.*, vol. 148, Feb. 2024, Art. no. 108889.

[14] T. Simons and D. J. Lee, "A review of binarized neural networks," *Electronics*, vol. 8, no. 6, pp. 661–687, Jun. 2019.

[15] E. D. Nardo and A. Ciaramella, "Tracking vision transformer with class and regression tokens," *Inf. Sci.*, vol. 619, pp. 276–287, Jan. 2023.

[16] S. Liu, P. Gao, Y. Li, W. Fu, and W. Ding, "Multi-modal fusion network with complementarity and importance for emotion recognition," *Inf. Sci.*, vol. 619, pp. 679–694, Jan. 2023.

[17] J. H. Kong, L. M. Ang, and K. P. Seng, "A comprehensive survey of modern symmetric cryptographic solutions for resource constrained environments," *J. Netw. Comput. Appl.*, vol. 49, pp. 15–50, Mar. 2017.

[18] M. Courbariaux, I. Hubara, D. Soudry, R. El-Yaniv, and Y. Bengio, "Binarized neural networks: Training deep neural networks with weights and activations constrained to +1 or -1," 2016, *arXiv:1602.02830*.

[19] M. Rastegari, V. Ordonez, J. Redmon, and A. Farhadi, "XNOR-net: Imagenet classification using binary convolutional neural networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2016, pp. 525–542.

[20] S. Zhou, Y. Wu, Z. Ni, X. Zhou, H. Wen, and Y. Zou, "DoReFa-net: Training low bitwidth convolutional neural networks with low bitwidth gradients," 2016, *arXiv:1606.06160*.

[21] A. G. Howard et al., "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[22] Z. Liu et al., "ReActNet: Towards precise binary neural network with generalized activation functions," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Berlin, Germany: Springer, Aug. 2020, pp. 143–159.

[23] X. Zhang, X. Zhou, M. Lin, and J. Sun, "ShuffleNet: An extremely efficient convolutional neural network for mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6848–6856.

[24] H. Qin, R. Gong, X. Liu, X. Bai, J. Song, and N. Sebe, "Binary neural networks: A survey," *Pattern Recognit.*, vol. 105, Sep. 2020, Art. no. 107281.

[25] R. Krishnamoorthi, "Quantizing deep convolutional networks for efficient inference: A whitepaper," 2018, *arXiv:1806.08342*.

[26] A. Bulat and G. Tzimiropoulos, "XNOR-Net++: Improved binary neural networks," 2019, *arXiv:1909.13863*.

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[27] P. Xue, Y. Lu, J. Chang, X. Wei, and Z. Wei, "IR^{2}Net: Information restriction and information recovery for accurate binary neural networks," *Neural Comput. Appl.*, vol. 35, no. 19, pp. 14449–14464, 2023.

[28] J. Ye, S. Zhang, T. Huang, and Y. Rui, "CDbin: Compact discriminative binary descriptor learned with efficient neural network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 3, pp. 862–874, Mar. 2020.

[29] R. Ding, T.-W. Chin, Z. Liu, and D. Marculescu, "Regularizing activation distribution for training binarized deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11408–11417.

[30] S. Leroux, B. Vankeirsbilck, T. Verbelen, P. Simoens, and B. Dhoedt, "Training binary neural networks with knowledge transfer," *Neurocomputing*, vol. 396, pp. 534–541, Jul. 2020.

[31] Z. Liu, B. Wu, W. Luo, X. Yang, W. Liu, and K. T. Cheng, "Bi-real net: Enhancing the performance of 1-bit CNNs with improved representational capability and advanced training algorithm," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 722–737.

[32] R. Gong et al., "Differentiable soft quantization: Bridging full-precision and low-bit neural networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4852–4861.

[33] H. Qin et al., "Forward and backward information retention for accurate binary neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2250–2259.

[34] H. Qin, X. Zhang, R. Gong, Y. Ding, Y. Xu, and X. Liu, "Distribution-sensitive information retention for accurate binary neural network," *Int. J. Comput. Vis.*, vol. 131, no. 1, pp. 26–47, 2023.

[35] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[36] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[37] N. M. Rodrigues, K. M. Malan, G. Ochoa, L. Vanneschi, and S. Silva, "Fitness landscape analysis of convolutional neural network architectures for image classification," *Inf. Sci.*, vol. 609, pp. 711–726, Sep. 2022.

[38] A. Krizhevsky, and G. Hinton, "Learning multiple layers of features from tiny images," Comput. Sci. Dept., Univ. Toronto, Toronto, ON, Canada, Jun. 2009.

[39] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[40] J. Bethge, C. Bartz, H. Yang, Y. Chen, and C. Meinel, "MeliusNet: Can binary neural networks achieve MobileNet-level accuracy?" 2020, *arXiv:2001.05936*.

[41] C. Liu et al., "RBCN: Rectified binary convolutional networks for enhancing the performance of 1-bit DCNNs," 2019, *arXiv:1908.07748*.

[42] J. Gu et al., "Bayesian optimized 1-bit CNNs," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4909–4917.

[43] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," 2018, *arXiv:1806.09055*.

[44] A. Bulat, B. Martinez, and G. Tzimiropoulos, "BATS: Binary architecture search," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 309–325.

[45] D. Kim, K. P. Singh, and J. Choi, "Learning architectures for binary networks," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 575–591.

[46] C. Hanlin et al., "Binarized neural architecture search for efficient object recognition," *Int. J. Comput. Vis.*, vol. 129, no. 2, pp. 501–516, Feb. 2021.

[47] C. Shi, Y. Hao, G. Li, and S. Xu, "EBNAS: Efficient binary network design for image classification via neural architecture search," *Eng. Appl. Artif. Intell.*, vol. 120, Apr. 2023, Art. no. 105845.

[48] J. Gu et al., "Projection convolutional neural networks for 1-bit CNNs via discrete back propagation," in *Proc. AAAI*, 2019, pp. 8344–8351.

[49] C. Liu et al., "Rectified binary convolutional networks for enhancing the performance of 1-bit DCNNs," in *Proc. 28th Int. Joint Conf. Artif. Intell.*, Aug. 2019, pp. 854–860.

[50] G. S. Xia, W. Yang, J. Delon, Y. Gousseau, H. Sun, and H. Maître, "Structural high-resolution satellite image indexing," in *Proc. ISPRS TC VII Symp.-100 Years ISPRS*, Vienna, Austria, Jul. 2010, pp. 298–303.

[51] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, San Jose, CA, USA, Nov. 2010, pp. 270–279.

[52] G.-S. Xia et al., "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[53] L. Bai, Q. Liu, C. Li, Z. Ye, M. Hui, and X. Jia, "Remote sensing image scene classification using multiscale feature fusion covariance network with octave convolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620214.

[54] J. Fang, Y. Yuan, X. Lu, and Y. Feng, "Robust space-frequency joint representation for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7492–7502, Oct. 2019.

[55] M. Liu, L. Jiao, X. Liu, L. Li, F. Liu, and S. Yang, "C-CNN: Contourlet convolutional neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 6, pp. 2636–2649, Jun. 2020.

[56] C. Shi, T. Wang, and L. Wang, "Branch feature fusion convolution network for remote sensing scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 5194–5210, 2020.

[57] Q. Wang, W. Huang, Z. Xiong, and X. Li, "Looking closer at the scene: Multiscale representation learning for remote sensing image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 4, pp. 1414–1428, Apr. 2022.



Weifeng Gao received the B.S. and Ph.D. degrees in applied mathematics from Xidian University, Xi'an, China, in 2008 and 2013, respectively.

He is currently a Professor with Xidian University. His research interests include optimization, evolutionary algorithms, and their applications in real world.

Prof. Gao was a recipient of the Hong Kong Scholar by the Mainland–Hong Kong Joint Postdoctoral Fellows Program, China, in 2017, and the One Hundred Talent Program by Shaanxi, China, in 2017.



Menghao Tan received the B.S. degree in information and computing sciences from Xi'an University of Technology, Xi'an, China, in 2022. He is currently pursuing the M.S. degree with the School of Mathematics and Statistics, Xidian University, Xi'an.

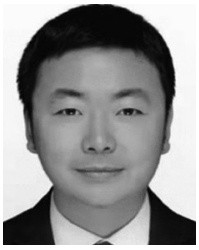
His research interests include evolutionary computation, neural architecture search, binary neural network, and computer vision, and their applications in real world.



Hong Li received the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2009.

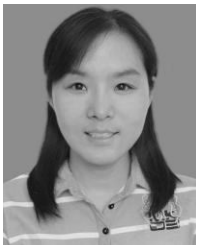
He is currently working as an Associate Professor with Xidian University. He has authored or coauthored more than 50 journals and conference papers in his research areas. His research interests include evolutionary computation, neural network, intelligent information processing, optimization theory, algorithms, and applications.

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Jin Xie (Member, IEEE) received the B.S. degree in mathematics from North China Electric Power University, Baoding, China, in 2013, and the Ph.D. degree in mathematics from Xidian University, Xi'an, China, in 2020.

He is currently a Lecturer with the School of Mathematics and Statistics, Xidian University. His research interests include machine learning, neural networks, semisupervised learning, and distributed learning.



Xiaoli Gao received the M.S. degree in applied mathematics from Xidian University, Xi'an, China. She is currently pursuing the Ph.D. degree with Sichuan University, Chengdu, China.

She is a Leader with Sichuan Jiuzhou Electrical Group Company Ltd., Mianyang, China. Her research interests include deep learning, convolutional neural network, radiation source, optimization, and their applications in real world.



Maoguo Gong (Fellow, IEEE) received the B.Eng. and Ph.D. degrees from Xidian University, Xi'an, China, in 2003 and 2009, respectively.

Since 2006, he has been a Teacher with Xidian University, where he was promoted to an Associate Professor and a Full Professor in 2008 and 2010, respectively, both with exceptive admission. He has authored over 100 journal articles and conference papers, and holds over 20 granted patents as the first inventor. He is leading or has completed over 20 projects as the Principle Investigator, funded by the National Natural Science Foundation of China and the National Key Research and Development Program of China. His research interests include computational intelligence, with applications to optimization, learning, data mining, and image understanding.

Dr. Gong was a recipient of the prestigious National Program for Support of the Leading Innovative Talents from the Central Organization Department of China, the Leading Innovative Talent in the Science and Technology from the Ministry of Science and Technology of China, the Excellent Young Scientist Foundation from the National Natural Science Foundation of China, the New Century Excellent Talent from the Ministry of Education of China, and the National Natural Science Award of China. He is the Executive Committee Member of the Chinese Association for Artificial Intelligence, a Senior Member of the Chinese Computer Federation, and an Associate Editor or an Editorial Board Member for over five journals, including IEEE TRANSACTIONS ON EVOLUTIONARY COMPUTATION and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.

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